I. Introduction to AI and Machine Learning Technologies

<u>What is Artificial Intelligence?</u>

There are many definitions of Artificial Intelligence (AI). We will use one of the most common: Artificial Intelligence is a set of techniques that enable software systems to perform tasks that we would expect to require human intelligence.[1] In other words, AI software *appears* to carry out cognitive tasks, such as:

- distinguishing pictures of different types of objects (e.g., identifying whether an image includes a cat or a dog, identifying the name of a person present in an image)
- recognizing speech (e.g., Amazon Alexa or Apple Siri accurately interpret a spoken command to provide a weather report or driving directions)
- translating languages (e.g., translating a paragraph of Spanish text to English)
- making decisions based on a visual input (e.g., deciding whether an image of a product being sold is high quality or low quality, determining whether video of articles on a factory assembly line shows a possible defect that should be removed)
- making decisions based on text inputs (e.g., determining whether text posted on social media reflects a positive or negative tone, determining whether a customer chat session with an online service representative should be escalated to a manager)
- generating realistic artwork from scratch (e.g., creating original images of paintings that are not simple copies or altered versions of existing paintings)
- generating realistic text (e.g., creating original paragraphs of a news article that is not a simple copy or altered version of existing text)
- playing a complex game (e.g., playing Go against professional human players, playing chess or Jeopardy!® against expert human players)
- developing a strategy that involves the decisions of third parties (e.g., planning global delivery routes to minimize transit costs and maximize certain benefits)
- provide a recommendation to a person based on preferences of that person and others (e.g., recommend a comedy movie that a person might enjoy given the movies that person has watched and movies liked by other comedy fans)

Note that this definition of AI does not require that the software actually mimic the cognitive process that humans would follow in performing these tasks. We only require that the *results* of the AI's task appear similar to the results a human would produce. In performing the task, AI software can (and most often does) utilize a very different approach than a human would take to accomplish

1 Some tasks might be adequately performed by non-human animals. For example, primates and some other animals can distinguish pictures of different types of objects and can understand spoken commands, tasks we typically associate with some modicum of "intelligence".

the same task. In fact, one could argue that the prominent and growing successes of AI applications today are the result of a deliberate strategy to avoid any use of actual “intelligence”.

What is machine learning?

The most vibrant area of AI work today is the subfield known as machine learning. Machine learning is a set of techniques that permit software to learn to perform a task without being explicitly programmed how to perform that task. Instead the machine learning software learns from *examples* of what it should do. In other words, machine learning involves showing software *what to do* but not *how to do it*. The process of learning from examples is called *training*, and the examples are usually called *training data* or a *training set*.

For example, assume that we would like a software system to recognize whether an email is spam or is not spam. An email system that recognizes spam could automatically place spam emails in a separate folder rather than annoy the user with unwanted emails in their primary inbox. To train such a machine learning system we could provide thousands of examples of email that are spam and thousands that are not spam. From these examples the machine learning learning system could learn to distinguish spam emails from non-spam emails. Specifically, from these examples the machine learning system would perform various types of sophisticated statistical analysis to learn which characteristics of an email are generally present in spam emails, and which characteristics are generally present in non-spam emails.

Perhaps in the set of examples (i.e., in the training data) the subject line of many spam emails contains all uppercase letters or exclamation points such as “YOU’VE WON!!!”, but very few non-spam emails have a subject line in all uppercase or with many exclamation points. The software would then learn that a subject line in all uppercase or with many exclamation points is associated with spam emails. Once the software has learned from these examples (i.e., once it has been *trained*), it would learn to predict whether a new email it processes has the characteristics of a spam email or a non-spam email.

This particular machine learning approach, known as *supervised machine learning*, allows the software to develop a model of the world induced solely from the observation of examples, and without being told *how* to perform the task. Machine learning therefore contrasts with an entirely different approach to AI in which a programmer provides the software with a set of rules that specify exactly how to recognize the pattern of interest. In the rules-based approach to creating a spam email detector, the software would have a set of rules that specify which features of an email (such as the subject line containing many exclamation points) suggest whether the email is spam.

One of the most significant disadvantages of the rules-based approach is that it requires a human to define a set of rules that precisely define exactly how the task will be performed. Not only is this laborious, it can be very difficult to do with high accuracy, especially for complex phenomena. For example, imagine you were required to create a set of rules that would allow software to determine whether an image was an image of US president George Washington and not an image of some other person. Even if you were capable of identifying George Washington images flawlessly, you might nevertheless be unable to specify exactly how you are able to do so. It might not be apparent exactly which parts of the image convey to you that it is or is not George Washington. You might be able to specify some helpful rules that you use to identify George Washington (e.g., he wears a gray wig, he never smiles, he wears colonial era clothing). Nevertheless these rules are likely to be incomplete because they provide a correct answer for some images but not all images. Perhaps they would fail when applied to an image of other males in colonial clothing who are not smiling.

Rather than attempt to explain *how* you would distinguish George Washington from other people, it would likely be much easier to simply use your ability to identify whether images contain George Washington. In other words, specifying *what* the software should do is easier than *how* the software should do it. You could for instance use a web browser to review a set of one hundred images of people, and click on the George Washington images. If a computer program were to capture this information from your browser, this would record your judgment about which of the images represented George Washington and which did not. Such a process by which a human applies their judgment to a set of images or other data is called *annotating* the data or *labeling* the data. A labeled dataset is simply data that includes a label, i.e., a desired decision, for each input. In the above example each input is an image and the desired decision is an indication of whether the image is of George Washington. Training data for this task could be a collection of, say, one hundred images, each including a 'yes' or 'no' to indicate whether it shows George Washington.

Machine learning, or the *data driven approach*, can address problems that involve far more complex phenomena than classifying spam emails. The more complex the problem, the harder it is to specify exactly *how* to do it with rules, and thus there is more of a relative advantage in simply specifying *what* should be done. Rules about how to perform a task, such as rules that specify how to recognize an image of George Washington, embody human judgment. A rules-based AI approach utilizes this human judgment by applying rules to inputs in order to produce outputs. Supervised machine learning also uses human judgment, though in the form of labeled data. During the training process the labels are used to teach the AI system to mimic the human judgment represented by the labels. Machine learning thus provides an easier way to provide human judgment to the software system.

A Framework for Understanding AI Systems

When assessing an existing or to-be-created AI system (also known as a *model*[2]) it is helpful to focus on a few high-level aspects:

The **general task** that the system should perform
The **specific outputs** that the system produces for this task
The **inputs** the system should use to produce its outputs
Which **type of machine learning** the system uses to learn to perform its task

The General Task

All AI systems produce some outputs intended to satisfy one or more goals, typically some sort of analysis or decision making. Most often the highest-level nature of these outputs accomplish some general task. Among the most common tasks are:

2 The use of the term "model" is especially popular in technical discussions. For example, it is common to refer to training a machine learning model, evaluating the outputs of the model, testing the model, and running the model on new inputs. It is helpful to remember that the use of the term "model" generally emphasizes that the AI system is merely an approximation that may or may not deliver results that are close to what we want. Different models can be similar but possess different strengths and weaknesses, perhaps excelling for some inputs but not others.

- *Classification*
- *Regression*
- *Content generation*
- *Clustering* or grouping inputs that are similar
- Formulating a *strategy* or set of steps

<u>Classification</u>

In classification the AI system must determine which of several categories an input belongs to. The previous example of spam detection is a classification problem; the system must determine whether each email (the input) belongs to the “spam” category or to the “not spam” category.

A classification task may involve any number of categories, not just two categories. For example, in image classification there may be thousands of categories each a possible object such as “car”, “boat”, “tree”, and “house”. As more classes are added it generally becomes harder to learn how to properly assign an input to the proper class. In addition, the more similar two classes are, the harder it is for the AI system to correctly distinguish them. For example, while it may be relatively easy to learn to distinguish airplanes from dogs, it will likely be harder to distinguish between two different dog breeds such as German Shepherds and Golden Retrievers.

<u>Regression</u>

In a regression task the AI system produces one or more numbers. The numbers may represent prices, distances, probabilities, economic data, measurements, or anything else that can assume a range of values. Unlike classification in which the output is one of some finite set of categories, in regression the output is a continuous value that can, in principle, assume any of an infinite number of values. For example, a model that predicts an optimal width of an aircraft engine could produce a value of 358, 358.1, 358.001, etc. Note that even if the number is confined to a finite range (e.g., between 0.0 and 1.0) there can still be an infinite number of values in that range.

<u>Content generation</u>

An AI system can create artificial but realistic data. The output can be text, such as blog posts, news articles, email subject lines, or marketing copy. The output can also be visual, such as artwork, photos, or video. This kind of AI task is often used to create realistic fakes, or “deep fakes”, resembling some true phenomena, such as a purported photo of a famous person.

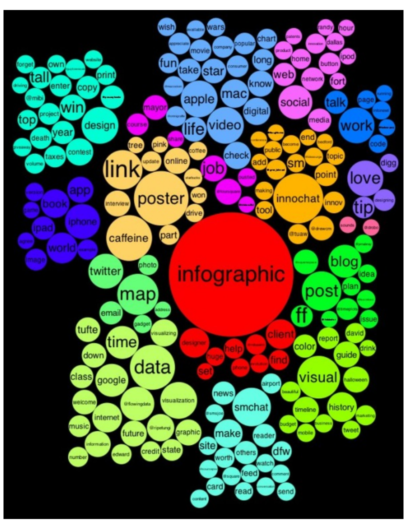
A content generation system is trained by providing it with numerous examples of the type of content that should be generated, whether it is Picasso paintings, Reuters news articles, or lines from python computer programs.

The process of content generation is most useful when it is guided, that is, when the output can be controlled to match specified preferences. For example, an AI system that automatically creates news text could be guided to create an article about a specific topic, such as a recent baseball game or an event in national politics.

Content generation most often results in original content, rather than simply copies or combinations of inputs previously seen by the AI system. The original artwork below is entirely computer generated, and was sold in 2018 by auction house Christie's:

Clustering

An AI system can be designed to cluster or group inputs based on how similar those inputs are to each other. For example, all tweets that appeared on Twitter last month could be clustered such that tweets related to the same topic appeared in the same group, or tweets with the same tone or style appear in the same group.



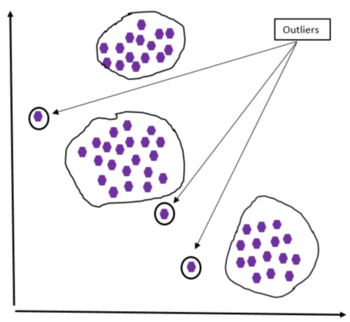
As another example, customers of a firm could be clustered according to customers’ revenue yield, product preferences, and / or service history.

Clustering is often used to create a convenient graph or chart to visually represent important information. For example, a chart of customer clusters could quickly convey to the viewer how many customers are in each cluster, or how similar or dissimilar various clusters are to each other.

In such charts the closeness of points represents their similarity, so points close to each other represent inputs (e.g., customers) that are similar.

Clustering is also used to make inferences about the inputs. Since inputs in the same cluster are considered similar to each other, one can infer that properties of one member of a cluster are likely to apply to other members of the cluster. For example, assume that customers of a firm are clustered based on their order history. If the firm learns that one customer has registered interest in a new product, it might be inferred that other customers in the same cluster might be interested in that new product.

Clustering is also used to detect anomalies or outliers. If the members of a cluster are assumed to be tightly concentrated, then inputs far away from all clusters can be considered outliers in the data set.



<u>Formulating a strategy</u>

Sometimes an AI system is designed to produce a strategy or best set of steps to take, rather than a simple output like a number or a classification. For example, it can be desirable to plan an optimal set of shipping routes, or an optimal configuration for a set of assembly line steps. AI systems can learn to produce a strategy for operating in a complex environment that maximizes or minimizes a specified objective, such as cost, revenue, or profit. AI systems can also learn to play nontrivial games such as chess or Pac-Man that require thinking ahead several steps to counter the anticipated moves of one or more opponents.

The Specific Outputs

Closely related to the general task of the AI system are the details of the outputs that are produced. For a regression task, this would include the details of what kinds of number or numbers are produced, what the numbers represent, whether the number is confined to some specific range. For a classification task, specific output details include how many categories, what each category represents, how exactly each category is defined, and whether the set of categories is comprehensive enough so that all inputs belong to some category. For content generation, the specific type or style of content to be generated (e.g., Wikipedia articles, images of kittens) must be specified. For clustering, we must outline precisely what features of the input are used to assess similarity and how exactly they are used in this assessment. For strategy formulation, critical output criteria include the possible steps or moves that can be taken in the strategy.

The Inputs

An AI system produces an output in response to one or more inputs the system receives. Understanding an AI system requires that we understand the exact nature of the inputs as well as what those inputs represent. The inputs can generally represent a variety of real-world phenomena, such as prices, measurements, other quantities, categories, images, text, video, speech, locations, or events. Combinations of different inputs are also possible. For example, a machine learning model in a medical application might output a patient's diagnosis based on inputs such as an image (an MRI of a patient's heart), text (doctor's notes from a previous evaluation), and categorical values (whether the patient's parents had a history of heart disease).

An important question is to assess the relation between the inputs and the outputs. Among the most significant issues is whether and why exactly we should expect the selected inputs to be sufficient for the output to be produced with acceptable accuracy. Stated another way, do the inputs contain enough information to produce the desired output. At one extreme is an input that has no connection whatsoever to the output. For example, if an AI system must predict an entering MBA student's final program GPA from their shoe size, we should not expect the output to be any more accurate than random guessing. At the other extreme, if an output can be completely determined by the inputs (i.e., the output is a deterministic function of only the inputs) then in principle a model can produce perfectly accurate outputs from these inputs. Most often the AI model must operate between these two extremes. Typically the AI model processes inputs that contain most but not all of the information necessary to provide an ideal output. Perhaps the output will always be close to the ideal output, but will necessarily contain some small but acceptable error.

Often the most challenging part of designing an AI system is to identify a wide range of inputs that collectively contain the information needed to produce an accurate output. This might require significant experience and expertise in the applicable business domain. For example, assume that an AI system must analyze customer chat sessions with a retailer in order to identify which customers are unsatisfied and in danger of switching to a competitor. An obvious input is the text of the chat sessions with the customer. However, other possibly relevant inputs include the history of orders of the customer, how many previous products that customer has returned, and how long that customer has been served by the retailer.

Four Types of Machine Learning

Four different types of machine learning are especially important.[3] Each type utilizes a different approach in learning to produce desired outputs or solve a class of problem.

- Unsupervised learning
- Supervised learning
- Self supervised learning
- Reinforcement learning

You can characterize each of the four types of machine learning as a spectrum based on how much supervision or guidance the machine learning system is given.

3 Keep in mind that different sources will divide the universe of machine learning techniques into more or fewer types. An unfortunate aspect of the machine learning field is that terminology can be used inconsistently by different authors, so terms such as "unsupervised learning" can be defined slightly differently.

Supervised learning
As described above, supervised learning involves training a machine learning system using labeled training data consisting of inputs (e.g., images) and the corresponding desired outputs (an indication of whether the image is of George Washington). The supervised learning process gradually enables the software to generate an appropriate output for different kinds of inputs it receives. The most common supervised learning tasks are classification and regression. Supervised learning is powerful due in part to the labeled data. The label data encapsulates human decision making about the inputs, and the training process seeks to endow the software with comparable decision making capabilities. The supervised machine learning approach, showing examples of *what* the software should do rather than telling software *how* to do it, is responsible for some of the most impressive advancements in AI over the past decade

Unsupervised learning
In unsupervised learning a model is trained with data that does not contain any labels. Therefore the training data does not contain any human assessment about how the inputs should be processed. For example, a system that clusters tweets by topic could be trained by simply providing it thousands of different tweets. The training data does not contain any indication of which tweets should be considered members of the same cluster. Instead, the unsupervised learning process uses algorithms to extract different features of the tweets (such as commonly occurring keywords) and use those features to group similar tweets.
Another use of unsupervised learning is to transform a data set of a large number of variables to a smaller number of variables. In doing this type of *data compression* (also known as *dimension reduction),* the software selects variables that offer a good approximation of the larger number of variables. Again, the training data contain no labels or human assessment of which variables should be kept or discarded. For example, in creating a model to predict consumer creditworthiness, each consumer might be defined by thousands of variables (such as all purchases, loans, and payments made over the customer's lifetime). To simplify the processing of millions of customers, an AI model could extract (or synthesize from existing variables) a smaller number of variables, such as the customer's savings balance, credit score history, and current debt-to-limit ratio. This compression from thousands of variables to just a few might nevertheless allow the model to accurately approximate each customer's current creditworthiness with far fewer computations.

Self-supervised learning
Although supervised learning is a powerful method for teaching a machine learning system how to replicate human judgment, that technique requires significant amounts of labeled data. It can be extremely time consuming and costly to create a sufficiently large labeled data set for a machine learning task. Consequently there are many tasks for which there are no labeled data sets available. To circumvent a lack of label data, self-supervised learning allows the AI system to generate labels from unlabeled data. These labels can then be used to train the system to be indirectly useful in performing other, related tasks.
The automatically created labels are not the kind that we would typically see in supervised learning. For example, the labels for use in a text processing application are commonly created by (1) providing thousands of paragraphs of text from documents, Wikipedia pages, etc., (2) randomly deleting a few words from each paragraph, and (3) using the deleted words as the labels. The

machine learning system would then be trained to predict the missing words in a paragraph – i.e., how to fill in the blanks.

It is not at all obvious why this sort of process would be beneficial. In short, an AI system that can predict missing words with high accuracy must more or less understand the surrounding text to some extent. Once the system is capable of “understanding” text, it can be leveraged with additional training to perform other text processing tasks that benefit from this capability. As one example, a system trained to accurately predict missing words in text can then be subsequently trained (with relative ease) to identify within the text of annual reports all statements related to future risks the company foresees.

Non-text data sets can be created in a similar manner. A self-supervised learning system could automatically create labels from a set of images by deleting random parts of the image and using the deleted parts as labels. Training the system to predict what the missing part looks like would endow the software with some understanding of images, or at least of the kinds of images in this data set.

More generally, the technique of self-supervised learning allows a machine learning system to understand some aspects of the structure of the data it is trained with, whether it is text, images, pricing data, or anything else.

This technique is unsupervised in that the original training data do not carry labels. However, the technique of automatically creating labels and then training with these labels is different enough that self-supervised learning should be considered distinct from unsupervised learning.

Self-supervised learning will be described in detail in a subsequent section.

<u>Reinforcement learning</u>

Reinforcement learning is a technique for learning strategies or processes that lead to optimal results in some rigorously defined environment. Often reinforcement learning is described in the context of teaching software how to play a video game at an expert level.

For example, the rules of Pac-Man are straightforward and completely defined. We can comprehensively specify what kinds of moves the player is allowed to make (the player can move through open parts of the maze but not through walls), what kinds of events happen at different times (when the player eats dots points are earned, when the player clears all dots the level has been successfully completed, if a ghost touches the player the player loses one life and play restarts).

Since the rules of the game are completely specified, we can allow the player to be controlled by AI software. Thus the AI software can play games repeatedly and without any human intervention whatsoever. As the AI software controls the player during game play, it receives feedback such as where the player’s Pac-Man icon is, whether the player is near or far from dots and the enemy ghosts, etc. This in turn allows the AI software to calculate its performance during game play, including the current score and whether the level has been successfully cleared. Therefore the AI software can determine what kind of performance is attained from different game playing strategies that the software executes.

The AI software operates analogously to how a human would: it plays the game many times, each time understanding how well it is playing, and which strategies generally lead to better scores. In the beginning the AI software employs randomized strategies. It gradually progresses to more effective and sophisticated strategies as it learns which strategies lead to better performance.

Non-game applications of reinforcement learning include different forms of resource management and planning in which the relevant ironment can be completely specified. Reinforcement learning will be described in greater detail in a subsequent section.

Training

Machine learning systems are trained using training data. Training most often involves the software iteratively trying to produce a “good” output by repeatedly producing outputs from the training data. With each iteration the software produces outputs and assesses how good the outputs are. The software then adjusts the manner in which it produces outputs from inputs in a strategic manner calculated to increase the accuracy in the next iteration. There may be hundreds, thousands, or millions of iterations depending on the amount of training data and the complexity of the task to be learned. In general, the more complex the problem, the more training data is required for the software to learn the necessary patterns that lead to good outputs.
Training can take a very long time (sometimes thousands of hours of computing time), may require costly computer hardware, and may even require tremendous amounts of electricity simply to run the computers performing the training. Some of the most complex state-of-the-art machine learning models are so complex that only a few companies can afford to train them.

Self Assessment Questions

You should spend enough time perusing this section to be able to answer the following questions.

What is the defining characteristic of machine learning?

How does machine learning differ from the rules-based approach to AI? What are the advantages and disadvantages of the rules-based approach as compared to machine learning?

Briefly describe the four types of machine learning and how the each differ from each other.

Apply the framework for understanding AI systems to the following goals:

> *In a company’s annual report, recognize and print the names of all company officers mentioned in the report*
>
> *Calculate the expected sales price of any single-family home in the city of Boston and its suburbs*
>
> *Determine whether a credit card purchase is possibly unauthorized by the card holder*
>
> *Predict the stock price for Amazon.com at the end of this year*
>
> *Assess whether a customer engaged in an online chat with a retailer’s AI chatbot should be transferred to a live agent*